

3 July 2026

Dear Editor,

On behalf of my co-authors, I am pleased to submit our manuscript, "*Quantifying superspreading in bacterial STI outbreaks using phylodynamics*", for consideration as a Research Article in *Nature Communications*.

*Neisseria gonorrhoeae* causes an estimated 82 million new infections every year, and rising resistance to the first-line antibiotics ceftriaxone and azithromycin has made prevention, rather than treatment, the priority. Like many sexually transmitted infections, its spread is highly heterogeneous: a small number of individuals — "superspreaders" — drive most onward transmission. How much superspreading occurs is decisive for choosing effective interventions, yet it is invisible to standard methods, which rely on the dense contact-tracing data that is rarely available for a largely asymptomatic infection.

Here we show that superspreading can be quantified directly from pathogen genomes alone. We developed a Bayesian phylodynamic model that partitions transmission between superspreaders and non-superspreaders, and paired it with a hierarchical partial-pooling strategy that shares information across outbreaks to overcome the limited genetic resolution of bacterial data. Key findings include:

- A multi-type birth–death model that estimates the fraction of superspreaders, their transmission advantage, and reproductive numbers with no contact-tracing data, released as an open-source BEAST2 package.
- In simulations calibrated to *N. gonorrhoeae* genomes, the model accurately recovers superspreading even under incomplete sampling and unknown case types.
- Applied to 18 gonorrhoea transmission clusters from Victoria, Australia (2019–2021), superspreading was widespread, with superspreaders transmitting on average ~30 times more than other cases — more extreme than the classic 20/80 rule.
- COVID-19 lockdowns lowered overall transmission, but superspreaders became the dominant driver of the clusters that kept growing — a mechanism that explains why population-level measures underperform when superspreading dominates, and why individual-level interventions such as targeted testing and partner notification matter.

To our knowledge, this is the first study to quantify superspreading in a bacterial STI from genomic data alone. Beyond *N. gonorrhoeae*, it provides a general, openly available framework for detecting transmission heterogeneity in pathogens where contact tracing is impractical, with direct relevance to STI control and to infectious-disease modelling more broadly. We believe these findings are sufficiently novel and of broad interest to warrant consideration for *Nature Communications*.

All authors have contributed to and approved the manuscript. The model is released as open-source software, and the data and code needed to reproduce our analyses are publicly available. The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere.

Thank you for considering our submission.

Sincerely,

Jordi Sevilla Fortuny, Julia Kende, Michael Meehan, and Sebastian Duchene